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Industry 4.0 as a data-driven paradigm: a systematic literature review on technologies

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Abstract

Purpose – The purpose of this paper is to identify current technologies related to Industry 4.0 and to develop a rationale to enhance the understanding of their functions within a data-driven paradigm.

Design/methodology/approach – A systematic literature review of 119 papers published in journals included in the Journal Citation Report (JCR) was conducted to identify Industry 4.0 technologies. A descriptive analysis characterizes the corpus, and a content analysis identifies the technologies.

Findings – The content analysis identified 111 technologies. These technologies perform four functions related to data: data generation and capture, data transmission, data conditioning, storage and processing and data application. The first three groups consist of enabling technologies and the fourth group of value-creating technologies. Results show that Industry 4.0 publications focus on enabling technologies that transmit and process data. Value-creating technologies, which apply data in order to develop new solutions, are still rare in the literature.

Research limitations/implications – The proposed framework serves as a structure for analysing the focus of publications over time, and enables the classification of new technologies as the paradigm evolves.

Practical implications – Because the technical side of the new production paradigm is complex and represents an evolving field, managers benefit from a simplified and data-driven approach. The proposed framework suggests that Industry 4.0 should be approached by looking at how data can create value and at what role each technology plays in this task.

Originality/value – The study makes a direct link between Industry 4.0 technologies and the key resource of this revolution, i.e. data. It provides a rationale that not only establishes relationships between technologies and data, but also highlights their roles as enablers or creators of value. Beyond showing the current focus of Industry 4.0 publications, this paper proposes a framework that is useful for tracking the evolution of the paradigm.

Keywords Digitization, Technology, Industry 4.0

Paper type Research paper

1. Introduction

A new paradigm is emerging in the manufacturing industry: digitized and networked production, based on a new and valuable resource, i.e. data. The term Industry 4.0, an *ex ante* allusion to the fourth industrial revolution, was coined in Germany to describe the phenomenon (Kagermann *et al.*, 2013; Lasi *et al.*, 2014). The expression was initially most commonly used in German-speaking countries but has recently been gaining popularity worldwide (Schneider, 2018), and scientists are beginning to use the term more consistently.

Despite its relevance, Industry 4.0 lacks any clear and concise definition (Piccarozzi *et al.*, 2018; Schneider, 2018). It is often characterized by its technology, or better, its technologies. Technology is a central element of the so-called industrial revolutions. Studies on the subject state that many of the economic and social changes that have taken place in society since the eighteenth century have been brought about by technology (Freeman and Soete, 1997; Mokyr, 1997; Landes, 2003). Perez (2002) attributes the periods of rapid development to technological revolutions. However, more important than technology *per se* is an understanding of what problem it tries to solve and how. March and Smith (1995) explain

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that "[t]echnology includes the many tools, techniques, materials and source of energy that humans have developed to achieve their goals" (p. 252). In this context, to understand the fourth industrial revolution, we should ask which technologies are involved and what goals they help humans to achieve. Managers must have a clearer view of the technical aspects of this paradigm and their roles in business.

Academics and practitioners attribute different technologies to Industry 4.0. It is possible to find studies that mention a dozen technologies or more (Chien *et al.*, 2017; Lin *et al.*, 2017; Liu *et al.*, 2017). Furthermore, the technologies mentioned do not always match. The absence of a reference to something that is part of this paradigm often leads to a conceptual panacea. At the same time, because it is a new phenomenon, the underlying concepts are still evolving, which means that new technologies may appear at any time. This situation makes it difficult to understand Industry 4.0 and, consequently, developing strategies related to it is problematic. Certainly, recent studies enhance understanding of the technologies associated with the new paradigm. Oztemel and Gursev (2018) present an extensive literature review and explain many technologies in detail. Chiarello *et al.* (2018) carry out a clustering exercise for 100 technologies using Wikipedia links, finding 11 technology clusters. Schlund and Baaij (2018) conduct a systematic literature review of 38 texts, assessing the importance of each technology and the frequency with which it is mentioned. Ghobakhloo (2018) identifies technologies and design principles and proposes a strategic roadmap toward Industry 4.0. These studies help to understand the relevance and developmental trends of Industry 4.0 technologies. However, we argue that there is still a need for a simplified and aggregated approach that can help in understanding the roles of current technologies and can be used to classify new ones. In this sense, the question that guides this study is:

RQ. How can Industry 4.0 technologies be organized into a framework, considering their functions relating to data and their role in this new paradigm?

The chosen approach for answering this research question is a systematic literature review of scientific papers to identify the technologies, combined with an examination of relevant texts to obtain the meaning of each technology. The result is a framework that explains the roles of technologies in the new production paradigm, serves as a structure to analyse the focus of publications over time and enables the classification of new technologies as the paradigm evolves.

The rest of this paper is organized as follows. In Section 2, we present Industry 4.0 definitions and characteristics, explaining data as the key resource of this revolution. Next, in Section 3, we describe the method and procedures used. In Section 4, we present the results of the systematic literature review, through descriptive and content analyses of the corpus. On the basis of these analyses, we develop and discuss the framework in Section 5. Finally, in Section 6, we present the study's conclusions.

2. Industry 4.0

The expression Industry 4.0 defines the next stage of industrial systems, where networked, automated and intelligent processes will be flexible and self-configurable (Schneider, 2018; Brettel *et al.*, 2014), improving efficiency and enabling the creation of new revenue sources (Dalenogare *et al.*, 2018; Oztemel and Gursev, 2018; Wang *et al.*, 2017). These processes will develop through a combination of digital and manufacturing technologies, which enable vertical integration of the organization's systems, horizontal integration in collaborative networks and end-to-end solutions across the value chain (Kagermann *et al.*, 2013).

Studies point out that in the analysis of Industry 4.0, it is more appropriate to include a set of technologies rather than each one individually (Posada *et al.*, 2015; UNIDO and Policy Links, 2017). However, it is not easy to identify which technologies are part of this paradigm because they are numerous, their names or components vary according to the specific use and they are still evolving (Chiarello *et al.*, 2018). To list the technologies of Industry 4.0 is a rough, imprecise and static way of defining it. At the same time, without understanding the underlying concepts of Industry 4.0, theoretical or practical advances become difficult. In this sense, framing the discussion is necessary. Here, we analyse Industry 4.0 as a data-driven paradigm because data are at the core of digitization (UNIDO and Policy Links, 2017; European Commission, 2014) and are the essential resource of the new industrial stage (Böhmer *et al.*, 2017; Scholz *et al.*, 2018).

2.1 Data as a key resource of Industry 4.0

In manufacturing, the use of collected data is often still limited to traceability (Mckinsey, 2014). In this context, data are of relative importance. The experience and knowledge of individuals or groups are what matters in decisions and actions. However, with the convergence of many technologies, more and better data are being captured, exchanged and analysed. This large quantity of data creates opportunities that go beyond monitoring and improving efficiency, promoting significant changes in manufacturing, organizations and society (Brynjolfsson and McAfee, 2014). The economy will be broadly affected by the use of data (OECD, 2017). Indeed, the European Union has made a direct comparison between data and an old and powerful resource, calling data the "new oil" (European Commission, 2014).

The technological revolutions of the past 200 years involved the widespread application of a key resource that was inexpensive and becoming cheaper; apparently inexhaustible; had potential application to various products, processes and sectors; and increased the power of capital and labour, reducing costs (Perez, 2002). Data have all these characteristics.

The third industrial revolution enabled the generation of large volumes of data, with considerable variety, speed and variability and decreasing costs (NIST Big Data Public Working Group, 2015). Sensors evolved significantly in terms of size, capacity, accuracy and cost, facilitating the generation and capture of data on things and people. The advent of the Internet promoted the connection between people and business, generating and communicating data at low cost (Chen *et al.*, 2012; McAfee and Brynjolfsson, 2012). Increased capacity and reduced processor costs, the possibility of remote and distributed storage and the emergence of free analytical software (such as R) and open-source platforms (such as Hadoop), also contributed to improving the relevance of data (Davenport *et al.*, 2012). The costs of using data are low and decreasing, so the first characteristic of a key resource, according to Perez (2002), is present.

Data are also an apparently inexhaustible resource. Unlike the critical resources of the previous industrial revolutions, data stocks do not reduce when used, and the quantity may even increase (when feeding simulations, resulting in new data, for example). Data can be used as many times as necessary and by several agents at the same time. This means they are non-rival and abundant goods (Brynjolfsson and McAfee, 2014). Thus, the second characteristic of a key resource is also present.

The third characteristic of a key resource, according to Perez (2002), is its extensive application. Indeed, data are currently used for a wide range of purposes: in the development of products (Davenport *et al.*, 2012; Porter and Heppelmann, 2014), in diagnoses of health problems (Chen *et al.*, 2012) and in precision agriculture (Porter and Heppelmann, 2014), among other purposes. This list shows only a small part of what is already being done by using data more intensively. There are many other areas where it is possible to apply this resource, including those with future potential related to artificial intelligence (AI) (Jordan and Mitchell, 2015).

Finally, the characteristic of increasing the power and reducing the cost of capital and labour is also present in the intensive use of data. Predictive methods, learning machines and self-configuring systems represent ways to do more with less. Increasing the efficiency of processes through the intensive use of data is one of the benefits attributed to Industry 4.0.

Industry 4.0 is a data-driven paradigm because it makes better use of data to create more value. Not surprisingly, many Industry 4.0 technologies are in some way related to data. Indeed, we have found two studies taking this approach. Lee *et al.* (2015) propose an architecture for Cyber-Physical Systems (CPS) in Industry 4.0 that shows this relationship on five levels: connection, conversion, cyber, cognition and configuration. Connection encompasses data acquisition and transmission. Conversion comprises processes to convert data into meaningful information. Cyber represents an information hub with analytics. Cognition is the use of information to support decisions. Configuration means that the production systems become self-adapting.

Working on a more direct relationship between data and technologies, Leal-Ayala and O'Sullivan propose a framework to explain the digitization of manufacturing (UNIDO and Policy Links, 2017). The authors use four groups of functions related to data to classify some technologies, namely, Data Generation and Capture, Data Transmission, Data Conditioning, Storage and Processing, and Data Application. The function of the first group is to provide the key resource for the manufacturing system. The second and third groups manipulate the data to ensure that they are in the right place at the right time, are in the correct form and have meaning. The fourth group represents the expected result of the process: the application of data to make things smart, facilitating flexible networked manufacturing systems. In Section 4.2.3, we use this framework, with some adaptations, to group the technologies that we identified through the systematic literature review, explained in detail in the next section.

3. Method and procedures

Following the guidelines of Budgen and Brereton (2006) and Kitchenham (2004), we conducted a systematic literature review in three stages: search, selection and analysis.

3.1 Search

For process documentation, we followed the recommendations of Vom Brocke *et al.* (2009) to guarantee the methodological rigour of the systematic review. We defined what, where, when and how to search for papers. Table I shows the parameters used.

Although expressions like smart manufacturing, manufacturing digitalization, and factory of the future are often used as synonyms for Industry 4.0, we chose to search only the terms Industry 4.0 and Industrie 4.0 (some documents written in English use the German word). This approach is justified because although alternative concepts may not differ significantly from Industry 4.0, they often address different levels of analysis, (i.e. industry, factory, processes), and have inaccurate meanings (Schneider, 2018). Additionally, some of these terms were used many years ago (see, e.g. Rosenthal, 1984; Schaffer, 1986; Welber, 1987; Meredith, 1987; Badham and Wilson, 1993; Huang and Zhang, 1995; Boyer *et al.*, 1996; Lee, 1998), even before the emergence of the central technology systems of Industry 4.0. This could be more confusing than informative. Furthermore, to enhance understanding of Industry 4.0, it makes sense to study the view of the scientific community using this term. The chosen

What	Where	When	How
"Industr* 4.0"	EBSCOhost	Up to July 2017	Fields: Title OR Abstract OR Subject Terms. Publication Type: Scholarly (Peer Reviewed) Journals/full texts
"Industr* 4.0"	Web of Science	Up to July 2017	Fields: Title OR Topic. PublicationType: Article
"Industry 4.0" OR "Industrie 4.0"	ScienceDirect	Up to July 2017	Fields: Title, Abstract, Keywords. Publication Type: Journals

Table I. Search parameters

expressions were searched for in the title, abstract and keywords fields, on the EBSCOhost portal and in the ScienceDirect and Web of Science databases. The period defined for the survey was from any date up to July 2017. We filtered the results to consider only papers written in the English language.

The search returned 97 articles from EBSCOhost, 276 from ScienceDirect and 226 from the Web of Science, giving a total of 599 texts. After an initial scan of the titles, 74 duplicates and one "call for papers" were excluded. From the 524 remaining papers, 210 (40 per cent) had been published in conference proceedings and 314 (60 per cent) in journals.

3.2 Selection and classification

To ensure the quality of texts, we selected only the articles published in journals included in the Journal Citation Report (JCR). Of the 314 journal papers, 177 (56 per cent) were published in journals included in the JCR. We then defined inclusion and exclusion criteria to include only texts that could contribute to the purpose of the review. Documents that only cited Industry 4.0 as a context, but did not discuss the paradigm and its technologies, were excluded. As the term originally referred to manufacturing, studies that dealt with the application of the concept outside the manufacturing value chain were also removed from the analysis. We included texts that were strongly related to the objective, that is, those that had, as their central theme, Industry 4.0 and its technologies, as well loosely related texts that defined Industry 4.0 and its technologies but focussed on other issues. This decision was made in order to increase the number of articles analysed. Table II shows the criteria used.

When the 177 abstracts were read, 12 were excluded by criterion EC1 and 2 by EC2. Of the 163 remaining articles, we did not have access to full texts in two cases. As a result, the corpus consists of 161 documents, that were uploaded to ATLAS.ti software and read in full. During reading, we excluded another 42 papers via EC1. The selection process resulted in 119 articles being included in the analysis according to IC1 (89) and IC2 (30). We created a memorandum with brief records of the contribution of each paper to the purpose of the review, and its classification according to the criteria. Figure 1 summarizes the procedures of the systematic literature review.

3.3 Analysis

We performed both descriptive and content analyses. The first uses the articles' metadata to characterize the corpus. The second identifies the technologies and their frequencies, then groups them and identifies relationships.

4. Analytical results

This section describes the systematic literature review results, presented via the descriptive and content analyses.

Criteria	Action	Code
Loosely related: paper is about Industry 4.0 but does not focus on technologies	Include	IC1
Strongly related: paper describes/explains Industry 4.0 technologies or solutions related to these technologies	Include	IC2
Unrelated: paper is not related to Industry 4.0 or uses the term only to contextualize or exemplify some fact	Exclude	EC1
Out of scope: paper uses the expression "Industry 4.0" in a broad sense, related to sectors other than the manufacturing sector	Exclude	EC2

Table II.
Inclusion and exclusion criteria

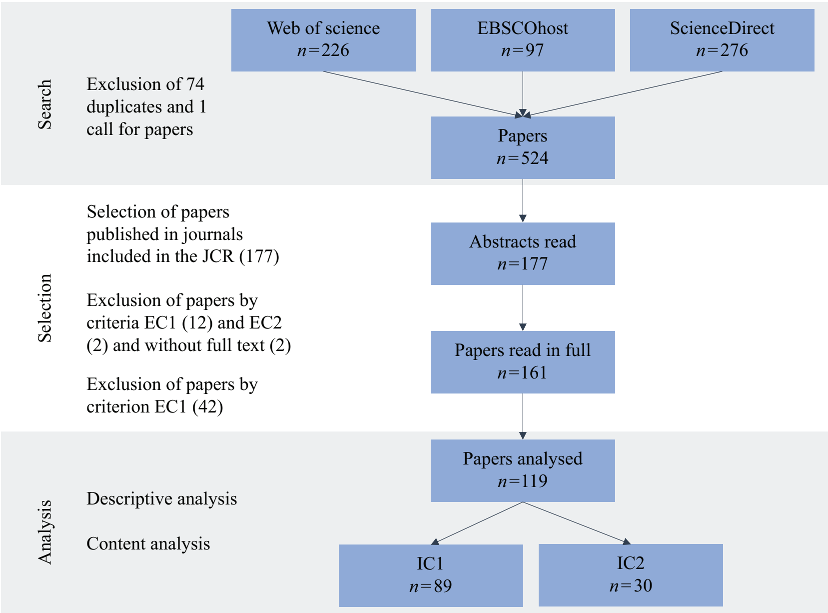

Figure 1.
Systematic literature review procedures

4.1 Descriptive analysis

The descriptive analysis of the 119 papers considers when and where they were published. Figure 2 presents the distribution of articles by year of publication.

Before 2015, there were no publications in JCR journals dealing with Industry 4.0 technologies. This can be explained by the novelty of the theme and by the fact that the expression was initially more commonly used in German-speaking countries (Lasi *et al.*, 2014). In 2016 and 2017, there were significant increases in the number of publications. This finding confirms the growing relevance of the term in the scientific community. Another fact reinforcing this idea is that 66 journals published the 119 papers, with about 75 per cent of journals publishing only 1 article and less than 8 per cent published more than 5. Figure 3 shows the journals that published more than two articles on the subject, together with their JCR indicators.

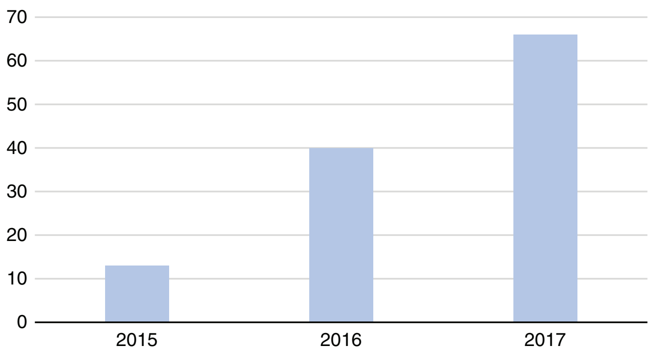

Figure 2.
Number of publications per year

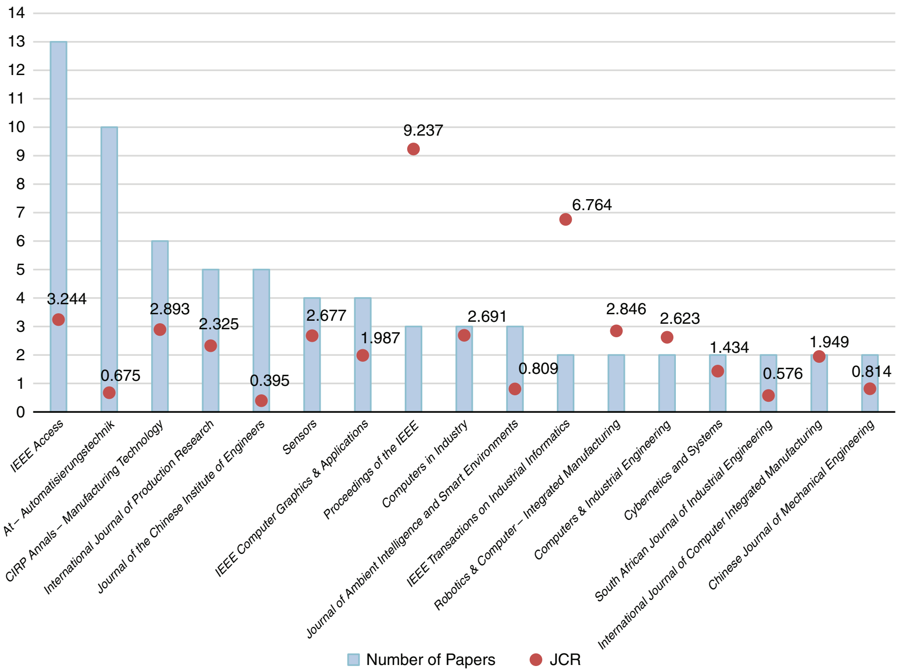

Figure 3.
Number of papers per journal and the journals' JCR indicators

The journals that published the most articles mentioning Industry 4.0 technologies are of a technical nature. We argue that this is related to the maturity level of the technologies. Because many of them are still in their infancy, there are still barriers to making the new paradigm feasible (Chen *et al.*, 2018). Several articles try to overcome current technical challenges in some technologies, such as spectrum inefficiency in large-scale wireless sensor networks, cybersecurity threats and lack of standards (see, e.g. Lin, Chen, Zhang, Guan and Shen, 2016; Kobara, 2016; Hortelano *et al.*, 2017).

4.2 Content analysis

Seeking to identify the technologies that are part of Industry 4.0, we read and codified the 119 papers. We concentrated on sentences that presented definitions of Industry 4.0, naming its technologies, or that cited technologies as part of the paradigm, without missing the whole message of each text. Examples of the structure of these sentences are as follows.

- Industry 4.0 is based on/is a combination of/has its foundations in/comes about with/consists of/encompasses/addresses [technologies].
- [Technologies] are the key for/are critical/central to/are cornerstones/essential parts of/play an important/central/pivotal/leading/crucial role/enable Industry 4.0.

When the technology appeared in sentences like those above, we marked the quotation and attributed a code to it. If the technology was only mentioned in passing or used to describe some specific application, we left it out. This choice is justified because our interest is in considering technologies that authors of the reviewed papers claimed to be part of Industry 4.0.

4.2.1 Identified technologies. In all, we identified 111 technologies. Each one was marked only once in each text, thus, representing its presence/absence in the paper. Table AI shows the codes and the numbers of papers in which each appears. To understand the codes, the following points must be borne in mind:

- technology is a concept that involves tools, techniques and materials or, more broadly, "a system of knowledge about how to manipulate nature in a logical scheme to achieve a functional transformation" (Betz, 2011, p. 333);
- terms such as digital technologies or information and communications technologies were not coded because they are too generic and do not provide answers to our questions;
- some technologies are not new and have been in use since the third industrial revolution, (e.g. Sensors and Actuators), but were still coded when linked to Industry 4.0 by the authors; and
- a code was generated for each expression cited by the authors, even if it was a synonym or a variation of a technology, (e.g. Cyber-Physical Production Systems and Industrial CPS) or only part of a technology (e.g. Big Data).

4.2.2 Most frequent technologies. Of the 111 technologies identified in the corpus, only a few appear more frequently. The top five technologies – CPS, Internet of Things (IoT), Big Data, Big Data Analytics and Cloud Computing – are analysed in more detail in this section. Table III shows their frequencies.

CPS represent the most frequently cited technology, confirming the position of the National Academy of Science and Engineering (acatech) and the German Research Centre for Artificial Intelligence (DFKI), which are considered central in Industry 4.0. Of the 119 articles, 77 cited CPS, some using the phrase as a synonym for Industry 4.0, as shown in Table IV.

The term CPS was coined in 2006 by Helen Gill, Director of the Embedded and Hybrid Systems Programme at the National Science Foundation, and it refers to a combination of physical and cybernetic systems (Lee, 2015). The two systems – the physical and the virtual – work as if they are one, so that everything that happens in the physical system impacts on the virtual system and vice versa (Lee, 2010). CPS involve the interaction of three elements:

Code	Number of texts	Percentage of the corpus
Cyber-Physical Systems	77	65
Internet of Things	70	59
Big Data	37	31
Big Data Analytics	33	28
Cloud Computing	27	23

Table III. Most frequent technologies

Paper	Quotation
Wang *et al.* (2016, p. 1)	The Industrie 4.0 describes a production oriented Cyber-Physical Systems (CPS) that integrate production facilities, warehousing systems, logistics, and even social requirements to establish the global value creation networks
Meudt *et al.* (2017, p. 413)	Industrie 4.0 (I4.0) promises nothing less than the 4th industrial revolution. Through cyber physical systems (CPS) a new level of organisation, control throughout value chains and complete lifecycle of products is expected to be achieved
Shafiq *et al.* (2015, p. 36)	Industry 4.0 is a powerful concept that promotes the computerization of traditional manufacturing plants and their ecosystems toward a connected and continuously available resources handling scheme through the use of Cyber-Physical Systems (CPS)

Table IV. Examples of quotations regarding Cyber-Physical Systems embedded systems, communication networks and their connection over the internet, and people (Geisberger *et al.*, 2011; NIST, 2013). Research on CPS is at an early stage, focussing on concepts, architecture and modelling, and still shows few actual cases of application (Baheti and Gill, 2011; Lee, 2015; Kang *et al.*, 2016).

CPS can be used in a wide range of industries (Geisberger *et al.*, 2011; Lee, 2015). In manufacturing, they involve all the physical elements – parts, components and finished products – that also have a virtual representation in the production systems (Kagermann *et al.*, 2013).

The second technology most often cited in the analysed papers is the IoT, which was present in 70 texts. Like CPS, the IoT is also often cited as a synonym for Industry 4.0, as shown in Table V. The term refers to the connection of an infinite number of "things" and objects in the network, so that each has its own identity and can be connected permanently or intermittently (Fenn and Lehong, 2011). Wang *et al.* (2017) state that "[. . .] the IoT can be regarded as a network where CPS interact with each other by unique addressing schemes" (p. 4).

Already glimpsing the evolution of the internet protocol version, the IoT concept was first documented in 2002 in a *Forbes Magazine* report. The concept has spread rapidly (Mattern and Floerkemeier, 2010). In 2011, Gartner included the IoT in its "Hype Cycle for Emerging Technologies" study. Like CPS, the IoT can be applied in a wide range of sectors. In manufacturing, its application allows machines to communicate with each other, and with the products (Atzori *et al.*, 2010).

The examples cited in Tables IV and V show that for some authors, CPS are the most representative of Industry 4.0, while, for others, the IoT has this distinction. Indeed, there are some overlaps between the definitions of these technologies in the literature, as discussed in Section 4.2.3.

The third and fourth technologies most cited in the articles have a very close relationship. These are Big Data (37) and Big Data Analytics (33). The Big Data concept represents the large quantity of data, structured and unstructured, from machines, products, components and people. Looking at this definition, Big Data is not *per se* a technology. However, as explained in Section 4.2.1, we created the codes according to the expressions cited by the authors, and some of them mention Big Data as a technology. Table VI contains examples of quotations.

Big Data is a relatively new topic in academia and the business world. For this reason, the concepts that guide it are still not entirely clear, nor are its boundaries well established (Chen *et al.*, 2012; Provost and Fawcett, 2013; Gandomi and Haider, 2015). Despite the lack of a universally accepted definition, the potential business impact has been widely discussed in recent years (Gandomi and Haider, 2015; Wamba *et al.*, 2015; Akter *et al.*, 2016; Davenport, 2014).

We believe that when authors cite Big Data as a technology, they are not only referring to massive volumes of data, but also to all the techniques and tools used to store, organize and process data, to transform them into valuable knowledge. In this sense, the expression has the same meaning as Big Data Analytics.

Table V.
Examples of quotations regarding the Internet of Things

Paper	Quotation
Chen and Tsai (2017, p. 131)	Industry 4.0 is the introduction of the Internet of Things and Services into the manufacturing environment
Sahay (2016, p. 1)	Industry 4.0, or Internet of Things (IoT), is an emerging technology trend in the manufacturing arena, and is widely considered the fourth industrial revolution
Lin, Deng, Chen, and Chen (2016, p. 47)	One of the foundations for Industry 4.0 is to integrate technologies of the Internet of Things (IoT) and smart manufacturing

Paper	Quotation
Wang et al. (2016, p. 3)	Needed Technologies. Some emerging information technologies, such as IoT, big data, and cloud computing as well as artificial intelligence (AI) technologies [. . .] are enabling factors of Industrie 4.0
Xu and Hua (2017, p. 17543)	An intelligent factory regards the Cyber-Physical Production System (CPPS) as its kernel and takes deep fusion of information technology and manufacturing technology as its characteristics that will embed the new generation of information technologies such as Internet of things, big data and cloud computing, into different segments of the manufacturing process [. . .]
Turner *et al.* (2016, p. 885)	Under Industry 4.0, advances in technologies ranging from cyber-physical systems (CPS) to the Internet of Things, Big Data, and Cloud technologies have been melded together into a comprehensive research agenda for the future of manufacturing

Table VI. Examples of quotations regarding Big Data

Big Data Analytics, as its name suggests, addresses the analysis of large quantities of data generated by connected equipment, machines, components, products and people. Analytics consists of using algorithms to find patterns and relevant information in the data. It uses statistical techniques as well as rules, heuristics and modelling (Provost and Fawcett, 2013). Table VII shows examples of sections encoded as Big Data Analytics.

Cloud Computing emerged as the fifth most frequently cited technology in the reviewed literature. NIST defines this technology as:

> [. . .] a model for enabling convenient, on-demand network access to a shared pool of configurable computing resources (e.g. networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction. (Badger *et al.*, 2012, pp. 2-1)

The definition highlights the relevance of the concept to Industry 4.0. The main reason is that decentralized decisions and self-configuring processes would require much investment in local assets and would not be practicable without the cloud. Some examples of quotations regarding Cloud Computing are shown in Table VIII.

The examples of quotations from the five most cited technologies show their relevance to Industry 4.0. However, this does not mean that other technologies are not critical. Indeed, the five most cited technologies are systems composed of, and related to, other identified technologies. Therefore, grouping these technologies by their functions related to data and establishing relationships between them may be helpful.

Paper	Quotation
Roy *et al.* (2016, p. 682)	Industry 4.0 provides an opportunity to collect more real time data about current state of machines, which can then be analysed using Big Data Analytics
Ang *et al.* (2017, p. 4)	[. . .] we summarized i4 as a collaborative network that combines seven key enabling technologies, namely intelligent robots, automated simulations, IoT, cloud computing, additive manufacturing, augmented reality and big data analytics
Edwards and Ramirez (2016, p. 110)	The term [Industrie 4.0] is being applied to a new phase in the digitization of manufacturing as a result of the growing use of sensors, the expansion of wireless communication and networks, the deployment of increasingly intelligent robots and use of artificial intelligence, the emergence of new forms of human–machine interaction through touch interfaces and augmented-reality systems as well as the development of “big data” analytics

Table VII. Examples of quotations regarding big data analytics

4.2.3 Grouping technologies. To group the technologies, we used as a reference the framework of Leal-Ayala and O'Sullivan, presented in UNIDO and Policy Links (2017). We assigned the technologies according to the functions they perform related to data, as follows:

- Group 1: Data Generation and Capture, which encompasses technologies that generate and collect data regarding products, components, machines, processes and people.
- Group 2: Data Transmission, composed of technologies related to the transport of data from their points of origin to the places where they will be stored or used.
- Group 3: Data Conditioning, Storage and Processing, consisting of technologies related to the guarding, maintenance, availability and transformation of data, creating knowledge.
- Group 4: Data Application, consisting of technologies that represent the application of data, thus changing the activities of the value chain.

The grouping proposed in this paper differs from Leal-Ayala and O'Sullivan's framework, especially regarding two technologies: IoT and CPS. In the previous framework, IoT is not classified in any group; it appears only in the title, which is "internet-of-Things simplified framework – an advanced manufacturing perspective" (UNIDO and Policy Links, 2017, p. 31). This indicates that Leal-Ayala and O'Sullivan use a broad definition of IoT. Here, we classified IoT in the Data Transmission group because we use the term IoT in a narrow sense to denote things connected to the internet, that is, as a communication technology. Indeed, there is no agreement in the literature about its definition (Atzori *et al.*, 2010, 2017). Atzori *et al.* (2017) argue that the descriptions of the concept range from too narrow to too broad, and reinforce the overlap with other fields of research, such as CPS. For the sake of clarity, despite its drawbacks, we have chosen the semantic concept of "a world-wide network of interconnected objects uniquely addressable, based on standard communication protocols" (INFSO D.4 Networked Enterprise and RFID *et al.*, 2008, p. 4).

Furthermore, in the structure presented in UNIDO and Policy Links (2017), CPS are classified as a data generation and capture technology, while, in this paper, we place CPS in the centre of the four groups, related to all functions. Technologies in Groups 1, 2 and 3 are components of, or used by, CPS, which, in turn, are applied to the technologies of Group 4. This approach is in line with the CPS concept in the reviewed papers, of which the following quotation is representative:

> CPSs are systems with embedded computers as a part of devices, buildings, medical systems, automotive systems, avionics, means of transport, etc. [. . .]. CPSs are able to: 1) capture physical data using sensors; 2) affect physical processes using actuators; 3) evaluate (and possibly save) captured data, and interact between the physical and virtual world; 4) are connected with local and/or global networks – wired and/or wireless. Furthermore, CPSs are cornerstones of the Industry 4.0 [. . .]. (Jirkovský *et al.*, 2017, p. 661)

Paper	Quotation
Wang *et al.* (2016, p. 3)	Some emerging information technologies, such as IoT, big data, and cloud computing as well as artificial intelligence (AI) technologies (e.g. MAS) are enabling factors of Industrie 4.0
Ma *et al.* (2017, p. 1)	Industry 4.0 is the fourth industrial revolution and describes the vision of a smart production system that meets market requirements using information communication technology (ICT), cyber-physical systems (CPS), Internet of things (IoT), and cloud computing
Preuveneers and Ilie-Zudor (2017, p. 292)	In that sense are cloud computing and Big Data critical enabling technologies for the Industry 4.0 paradigm

Table VIII.
Examples of quotations regarding cloud computing

Figure 4 shows the structure we used to group technologies.

We argue that this structure is appropriate because we found, in the corpus, definitions of Industry 4.0 that mention these four functions related to data. Many of them mention more than one, as shown by the quotations in Table IX (our underlining).

Grouping and establishing relationships between technologies based only on 119 papers was not possible, since most articles only cite them, without explaining their meaning or the links between them. Therefore, we also used other bibliographies.

Of the 111 technologies presented in Table AI, 3 refer to CPS and were not classified. We also did not group 44 technologies cited only once in the corpus, because their frequency suggests less relevance to the paradigm. Table X shows the remaining 64 technologies, classified into 4 groups.

The number of technologies per group indicates what has been considered most relevant in the research carried out. Group 2 has the largest number of codes (23). Only slightly behind

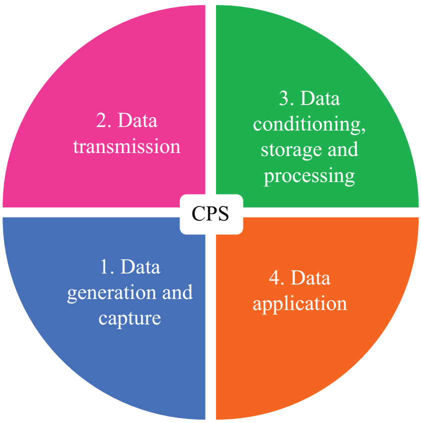

Figure 4. Structure for technology grouping

Table IX. Examples of Industry 4.0 definitions mentioning data functions

Paper	Quotation	Functions
Lin *et al.* (2017, p. 5)	IoT and CPS are what at stake in the fourth industrial revolution. The smart factory or smart something must build on the fitting of data processing, communication, and interaction between machines	Data Application Data Conditioning, Storage and Processing Data Transmission
Chien *et al.* (2017, p. 553)	CPS that integrates computation and physical processes is one of the main components of Industry 4.0. In particular, CPS consists of embedded computers and smart sensor networks to monitor, collect real-time data, and thus control the physical processes, usually with feedback loops where physical processes affect computations and vice versa	Data Generation and Capture Data Transmission Data Conditioning, Storage and Processing
Lobo (2016, p. 18)	The Industry 4.0 model is inherently a de-centralized one with masses of data being transferred. [. . .] Using the Industrial Internet of Things (IIoT), products will have the ability to collect and transmit data; communicate with equipment, and take intelligent routing decisions without the need for operator intervention. Cloud computing technology further gives a ready platform to store this data and make it freely available to systems surrounding it	Data Generation and Capture Data Transmission Data Conditioning, Storage and Processing

1. Data Generation and Capture 6 technologies	2. Data Transmission 23 technologies	3. Data Conditioning, Storage and Processing 22 technologies	4. Data Application 13 technologies
Sensors (26)	Internet of Things (70)	Big Data (37)	Smart Factories (13)
Embedded Systems (23)	Internet of Services (20)	Big Data Analytics (33)	Smart Manufacturing (9)
Actuators (14)	Internet (17)	Cloud Computing (27)	Robotics (6)
RFID (7)	Networks (15)	Artificial Intelligence (15)	Smart Products (5)
Smart Sensors (3)	Standards and Protocols (13)	Cloud (15)	Autonomous Vehicles (4)
Advanced Sensors (2)	Industrial internet of Things (10)	Cybersecurity (13)	Smart Machines (3)
	Semantics and Ontologies (10)	Augmented Reality (9)	Smart Robots (3)
	Wireless Sensor Networks (10)	3D Printing (7)	Smart Services (3)
	Communication Networks (6)	Additive Manufacturing (5)	Smart Artifacts (3)
	Industrial Wireless Networks (6)	Simulation (5)	Autonomous Robotics (2)
	Machine Networks (6)	Virtual Reality (4)	Flexible Manufacturing Systems (2)
	Service-Oriented Architecture (6)	Cloud Manufacturing (4)	Smart Grid (2)
	Machine to Machine (5)	Industrial Big Data (4)	Wearables (2)
	Mobile internet (5)	Advanced Analytics (4)	
	Computer Networks (4)	Agents (3)	
	Sensor Networks (4)	Blockchain (3)	
	Wireless Networks (4)	Interface Technologies (3)	
	Industrial Networks (3)	Visual Computing (3)	
	Internet of Everything (3)	Digital-to-Physical Transfer (2)	
	Horizontal and Vertical System Int. (2)	Human-Machine Interface (2)	
	Digital Networks (2)	Industrial Cloud (2)	
	Industrial Wireless Sensor Networks (2)	Machine Learning (2)	
	Wired Networks (2)		

Table X.
Grouped technologies

is Group 3, with 22 technologies. The total frequency of codes in each group confirms the relevance.

With only 6 codes, Group 1 technologies appear in 75 articles, while the 13 technologies associated with Group 4 appear in 57 articles.

4.2.4 Relating technologies. As there are many technologies in each group, we developed a network to understand better the relationships between them, as shown in Figure 5. Six types of links were identified, namely, “applied to”, “enables”, “is a”, “is associated with”, “is part of” and “uses”. The first finding is that the technologies are indeed very closely related. This does not mean that one cannot implement stand-alone technologies but that, in terms of Industry 4.0, they are interconnected.

The second finding is that a variety of terms can have the same meaning, or slight variations of meaning, in the area of application. One can see this by considering the most commonly used link, “is a”, with 36 occurrences. This suggests the absence of a standard terminology.

The third point to highlight is that many technologies are components of others, as indicated by the 21 “is part of” links. This means that many technologies are, in fact, technology systems and subsystems.

5. Discussion

Industry 4.0 is a new paradigm driven by a new valuable resource, i.e. data. It is often described by many technologies in a non-systematic way. Drawing on a systematic literature review and guided by the concept of a data-driven paradigm, we identified Industry 4.0 technologies, counted their frequencies and grouped and related them. In this section, we discuss the results and contributions of the study.

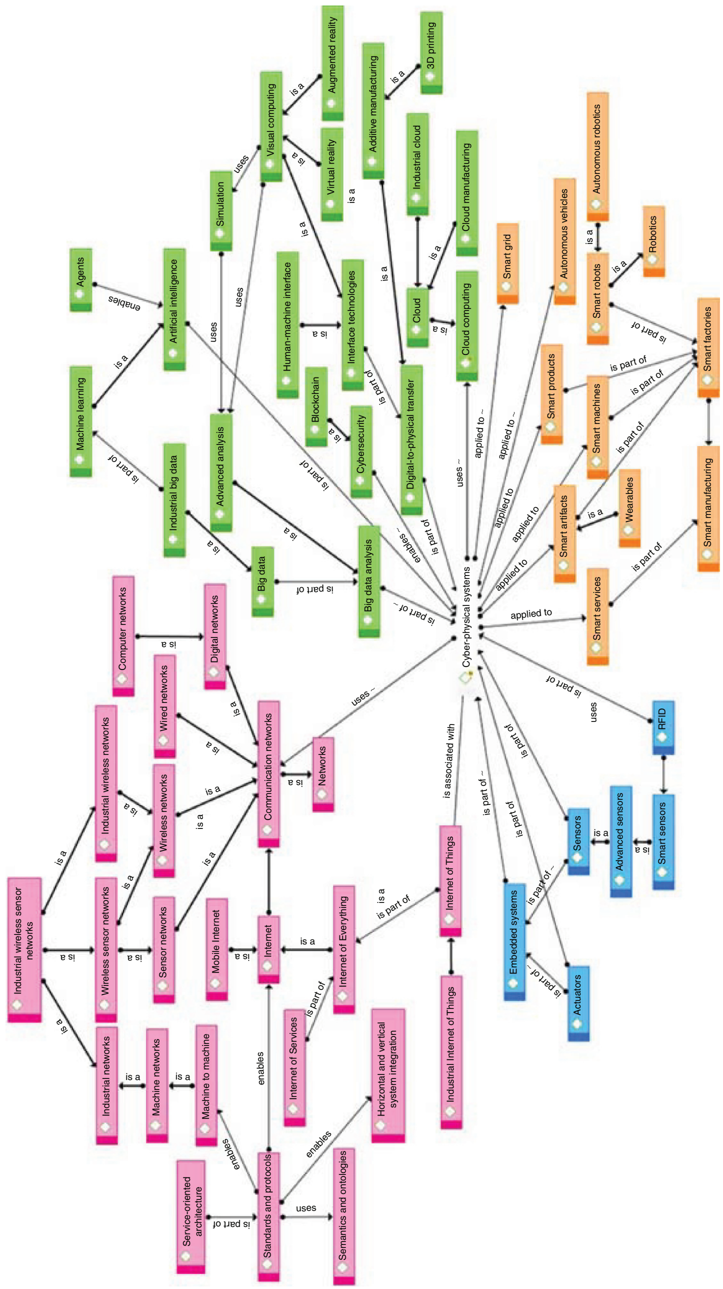

Figure 5.
Relationships between technologies

The number of technologies identified in the corpus is high. This is because the paradigm encompasses a constellation of techniques. As asserted by Posada *et al.* (2015) and stated in UNIDO and Policy Links (2017), it makes sense to analyse Industry 4.0 by looking at a set of technologies instead of each one individually. At the same time, we cannot attribute the same value to them all, because they have different frequencies and functions.

Out of 111 technologies, a few are often cited. The five most frequently cited are CPS, the IoT, Big Data, Big Data Analytics and Cloud Computing. These technologies are systems composed of, and related to, many other technologies, as can be seen in Figure 5. In fact, rather than five technologies, we can say that four emerged as the core of Industry 4.0, because, as we mentioned in Section 4.2.2, the term Big Data does not represent a technology *per se*, and the reviewed papers use the term in a broader sense than that of the technical meaning. For this reason, from now on we will refer to Big Data and Big Data Analytics as one technology.

These four technologies also emerged as relevant to Industry 4.0 in the studies of Schlund and Baaij (2018), Oztemel and Gursev (2018) and Ghobakhloo (2018). In addition to the high frequency of occurrence of these technologies in the corpus, other terms related to them also appeared in the reviewed literature. For CPS, some authors also use the terms Cyber-Physical Production Systems and Industrial CPS, which establish the application domain of CPS. In the same vein, rather than the IoT, some papers use the term Industrial IoT, (referring to the use of IoT in an industrial setting), Internet of Services (which specifies the "things" as services) and Internet of Everything (which is a broader concept including data and people as well as things). Related to Big Data Analytics, we also found the terms Industrial Big Data, which also refers to the application domain, and Advanced Analytics. For Cloud Computing, there are many expressions in the corpus, including the broader definition Cloud, domain definitions such as Industrial Cloud or Cloud Manufacturing and definitions that include the word "platform", indicating the environment where processing takes place, such as Cloud Platforms and Cloud Computing Platform.

Although technologies mentioning "networks" are not among the most frequently cited technologies, when the frequency of all types of network technologies is considered, (i.e. Broadband Networks, Cellular Networks, Communication Networks, Computer Networks, Digital Networks, Industrial Networks, Industrial Wireless Networks, Industrial Wireless Sensor Networks, Machine Networks, Mobile Communication Networks, Networks, Sensor Networks, Smart Sensor Networks, Software-Defined Networks, Wired Networks, Wireless Networks and Wireless Sensor Networks), we find these are cited in almost 50 per cent of the texts. This indicates that technologies that connect devices over cables or wirelessly are relevant in Industry 4.0.

A look at the 44 technologies cited only once is also revealing. On the one hand, this happens because authors interpret Industry 4.0 too broadly, including every emerging technology as part of the concept. Lin *et al.* (2017), for example, cite Agritech, Fintech and Electric Vehicles, and Chung and Kim (2016) cite Biotechnology, Nanotechnology and New Materials, among others. We agree that these technologies will have significant effects on society. Nevertheless, it is difficult to fit them into the narrower definition of Industry 4.0, which focusses on digitized and networked production (Kagermann *et al.*, 2013). On the other hand, some technologies, like Deep Learning, Data Mining and Automated Simulation, were mentioned only once but are, in fact, variations, subgroups or parts of other systems. They are therefore part of the paradigm. Furthermore, one must consider that technologies with low frequency of occurrence in the corpus, such as Blockchain technology, for example, are becoming more relevant. This means that Industry 4.0 is a dynamic concept, and whatever structure we use to analyse it, we must remember that these frequencies will change, and new technologies will emerge (Chiarello *et al.*, 2018).

Industry 4.0 combines mature and emergent technologies that can be applied in many domains (Chiarello *et al.*, 2018). Some variations in the expressions used to define it are to be expected. However, one must be careful because neologisms often hamper communication. In this sense, grouping technologies helps us to understand the field. Analysing the groups provides some valuable insights into this data-driven paradigm.

Data Transmission is the most populated group regarding the variety of technologies (23) and frequency of codes (225). Only slightly behind is the Data Conditioning, Storage and Processing group, with 22 codes marked 202 times. This is consistent with the characteristics that define Industry 4.0: vertical, horizontal and end-to-end integration (Kagermann *et al.*, 2013). In order to integrate, it is essential to establish data flows between the participants in the value chain – not only raw data but also analysed data, bringing knowledge to the system. Wireless communication, new internet protocols, remote and distributed storage and new analytics techniques are some of the recent advances that make this possible (Davenport *et al.*, 2012). Of course, there are still challenges to be overcome (Chen *et al.*, 2018) which require research and lead many publications to focus on these technologies.

The two groups with fewer codes behave differently. The Data Generation and Capture group contains the lowest number of technologies (six), yet these are present in more texts than the technologies in the Data Application group. We argue that, due to the higher level of maturity of technologies that capture and generate data, there are fewer variations in the names, and for this reason only six technologies were identified. Indeed, these terms have been used for many decades in manufacturing (Mckinsey, 2014). Conversely, Data Application technologies are predominantly new, (e.g. Smart Manufacturing, Autonomous Vehicles, Wearables). They are systems composed of multiple techniques, and because they are still in their infancy, they occur comparatively rarely in the publications.

Moreover, the four data functions are present in the concept of Industry 4.0 and, as Figure 5 shows, they are related through CPS. Technologies in Groups 1, 2 and 3 are components of, or are used by, CPS, which are applied to the technologies of Group 4. The relationship network makes it clear that, when implementing Industry 4.0, technologies from all groups will necessarily be present. However, as technologies in Group 4 represent the application of data to change the activities of the value chain, they are more directly linked to value creation. It is evident that the technologies that apply data are only feasible when the data is in the right place and form, with the correct meaning, and at the right time. Therefore, we propose new grouping: technologies that generate, capture, transmit, condition, store and process data, and are called Enabling Technologies. Although necessary, they are not enough to bring about the transformation claimed in the literature. Technologies that apply data are Value-Creating Technologies because they realize the potential of Industry 4.0. Figure 6 shows the framework for Industry 4.0 as a data-driven paradigm.

Logically, for value-creating technologies to occur, it is necessary to develop the enablers. For this reason, in the analysed publications, there is a higher frequency of codes in Groups 2 and 3. Industry 4.0 is an *ex ante* definition of an industrial revolution, so scholars are still trying to overcome the technical challenges of these groups. However, as technical barriers disappear, topics involving value-creating technologies, such as business models and value chain transformation, become essential.

The contributions of the developed framework are threefold. First, it simplifies a populated and evolving technology field, through the understanding of Industry 4.0 as a data-driven paradigm. Second, it serves as a structure for analysing the focus of publications over time. Third, it enables the introduction of new technologies as the paradigm evolves.

The technical features of Industry 4.0 are complex and changing. The number of technologies one can find in official publications, scientific papers or business magazines is high. As a result, research into how to manage the transition to Industry 4.0 takes place slowly. At the same time, companies do not know how to begin the transition. With this in mind, we proposed a simplified but still comprehensive way to aggregate technologies and enhance the understanding of their roles in the paradigm. The framework shows that

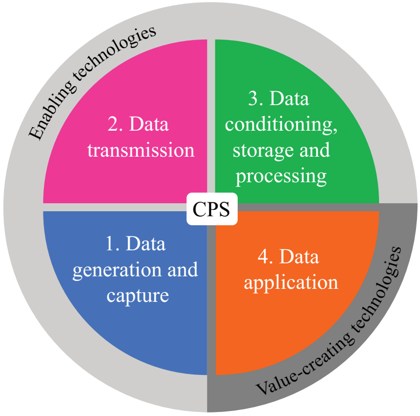

Figure 6.
Framework for Industry 4.0 as a data-driven paradigm

Industry 4.0 aims to make better use of data to create value. The first insight from this approach is that some technologies enable value creation, putting the data in the right form, in the right place, at the right time and with the correct meaning. Some create value by applying the data to improve a product or a process. This suggests that companies should begin their digitization process by asking themselves how, in their business, data can be applied to generate value, and then choosing the technologies to enable it. Data, not technology, will drive the transformation (Vanauer *et al.*, 2015; Mckinsey, 2018). The second insight is that a combination of technologies is necessary. This is relevant because many players are selling stand-alone enabling technologies, claiming Industry 4.0 benefits. Managers must be aware of what each technology can do and avoid exaggerated expectations. Sometimes, the complexity of phenomena and the pressure to embrace new technologies can lead to (expensive) disappointments. The central questions, when thinking about Industry 4.0 technologies are: "How can data be applied to my business to generate value?", and "In which technologies should I invest to enable this?"

The framework also serves as a structure to analyse the focus of publications over time, present what is being researched and suggest where research is needed. The analyses performed in this paper show that the technologies of Groups 2 and 3 are more frequently mentioned in the reviewed articles. However, this does not mean that these are the most important functions of the paradigm. They are the most debated currently because they still pose challenges. Group 1 contains more mature technologies, while Group 4 encompasses technologies that are just emerging. Not surprisingly, because of the challenges of Groups 2 and 3, there are still only a few cases of solutions using data to achieve vertical, horizontal and end-to-end integration. Consequently, fewer studies are citing them.

Finally, the third benefit of this framework is that it does not limit the number of technologies that can be analysed, enabling the classification of new technologies as the paradigm evolves. This is an essential feature because Industry 4.0 is still in its infancy, which means that new technical solutions will emerge (Kagermann *et al.*, 2013; Chiarello *et al.*, 2018). To classify a technology, one must question the role it plays concerning data, and then establish links with existing technologies.

6. Conclusions

This paper presents Industry 4.0 as a data-driven paradigm because data are the key resource of the new manufacturing transformation. Through a systematic literature review, it identifies 111 technologies. The findings confirm that Industry 4.0 is a complex subject, requiring a clear and concise definition (Piccarozzi *et al.*, 2018; Schneider, 2018) and that mapping its technology is not easy (Chiarello *et al.*, 2018). By grouping and linking identified technologies, a framework is developed to explain that they have different functions related to data. The structure shows three groups of enabling technologies and one group of value-creating technologies. CPS, which are central in the Industry 4.0 concept, are composed of, or use, technologies from the first three groups, and are applied to the technologies of the fourth group.

The framework enhances understanding of Industry 4.0 as a data-driven paradigm, helping companies to analyse the technologies according to their role. It also serves as a structure to follow the focus of publications over time and enables the classification of new technologies as the paradigm evolves. Because the technical side of this revolution is complex and represents an evolving field, scholars and managers will benefit from a simplified and data-driven approach. On the practical side, the structure suggests that one must approach Industry 4.0 by looking at how data can create value and what role each technology plays in this task. On the theoretical side, the structure offers a rationale for tracking the evolution of technologies linked to Industry 4.0 and identifying research opportunities. To test the usability of the framework, we suggest using it in companies that are planning to implement, or are already implementing, Industry 4.0 technologies, then assessing how the structure has helped them.

Furthermore, based on the results of the systematic literature review conducted in this paper, we also suggest the following:

- Intensifying the research into Data Application technologies, which is the group that creates value.
- Exploring the relationships between technologies and groups using Social Network Analysis to see how they connect, to determine the density of connections, and to identify which is the most central. This analysis would help understanding of how they enable and/or complement each other.

Of course, as always, creating a framework means leaving some things out. In this article, because of the nature of the reviewed publications, the results were in terms of "hard" technologies. However, "soft" technologies, such as management processes and organizational forms, are also essential to the success of Industry 4.0. In this sense, future research could use the framework to analyse how these technologies enable and create value for businesses.

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Appendix

Code	n	Code	n
Cyber-Physical Systems	77	Horizontal and Vertical System Integration	2
Internet of Things	70	Smart Grid	2
Big Data	37	Wearables	2
Big Data Analytics	33	Digital-to-Physical Transfer	2
Cloud Computing	27	Human-Machine Interface	2
Sensors	26	Industrial Cloud	2
Embedded Systems	23	Machine Learning	2
Cyber-Physical Production Systems	20	Advanced Sensors	2
Internet of Services	20	Digital Networks	2
Internet	17	Industrial Wireless Sensor Networks	2
Artificial Intelligence	15	Wired Networks	2
Cloud	15	Agritech	1
Networks	15	Biotechnology	1
Actuators	14	Energy Storage	1
Cybersecurity	13	Fintech	1
Standards and Protocols	13	Nanotechnology	1
Smart Factories	13	New Materials	1
Industrial Internet of Things	10	Presetters	1
Semantics and Ontologies	10	Quantum Computing	1
Wireless Sensor Networks	10	Advanced Robots	1
Augmented Reality	9	Automated Simulation	1
Smart Manufacturing	9	Drones	1
3D Printing	7	Electric Vehicles	1
RFID	7	Grid	1
Robotics	6	Reconfigurable Machine Control System	1
Communication Networks	6	Smart Cities	1
Industrial Wireless Networks	6	Smart Components	1
Machine Networks	6	Smart Design	1
Service-Oriented Architecture	6	Smart Enterprises	1
Additive Manufacturing	5	Smart Innovation	1
Smart Products	5	Smart Materials	1
Machine to Machine	5	Smart Procedures	1
Mobile Internet	5	Smart Processes	1
Simulation	5	Smart Supply Chain	1
Autonomous Vehicles	4	Cloud Computing Platform	1
Virtual Reality	4	Cloud Platforms	1
Cloud Manufacturing	4	Community Platforms	1
Industrial Big Data	4	Data Mining	1
Computer Networks	4	Deep Learning	1
Sensor Networks	4	Fog Computing	1
Wireless Networks	4	Machine Control Software	1
Advanced Analytics	4	Ubiquitous Computation	1
Industrial Cyber-Physical Systems	3	Visual Analytics	1
Smart Machines	3	Data Acquisition Systems	1
Smart Robots	3	Microcontrollers	1
Smart Services	3	Mobile Computing	1
Agents	3	Near Field Communication	1
Blockchain	3	Broadband Networks	1
Interface Technologies	3	Cellular Networks	1
Visual Computing	3	Industrial Ethernet	1
Smart Sensors	3	Industrial Internet	1
Industrial Networks	3	Mobile Communication Networks	1
Internet of Everything	3	Semantic Web	1
Smart Artifacts	3	Smart Sensor Networks	1
Autonomous Robotics	2	Software-Defined Networks	1
Flexible Manufacturing Systems	2		

Table AI. Identified technologies and their frequency in the corpus

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